Product Summary

V_{DS}	$R_{DS(ON)_TYP}$	I_{D_MAX}
40 V	2.0 mΩ @V_{GS} = 10V	146 A
	2.9 mΩ @V_{GS} = 4.5V	

Sinesemi Automotive MOSFET

AEC Qualified RoHS HALOGEN FREE

Features

- N-Channel Enhancement Mode - Logic Level
- AEC-Q101 Qualified, PPAP Capable
- 175°C Operating Temperature
- 100% ΔVDS & UIS & Rg Tested

Application

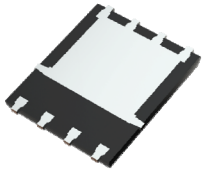
- General Automotive Applications

Mechanical Data

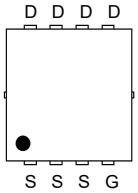
- Green Molding Compound
- Moisture Sensitivity: Level 1 per J-STD-020
- UL Flammability Classification Rating 94V-0

PDFN5060-8L

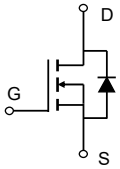
Top View



Bottom View



PIN Configuration
(Top View)



Schematic Diagram

Ordering Information

Orderable Part Number	Package Type	Device Marking	Form	Quantity (pcs)
SMT4003ALPQ	PDFN5060-8L	4003ALQ	13" Tape&Reel	5,000

Maximum Ratings (@ T_C = 25°C, unless otherwise specified.)

Parameter		Symbol	Value	Unit
Drain - Source Voltage		V_{DS}	40	V
Gate - Source Voltage		V_{GS}	±20	V
Continuous Drain Current (V_{GS} = 10V) (1)	T_C = 25°C	I_D	146	A
	T_C = 100°C		103	A
Pulsed Drain Current (2)		I_{DM}	584	A
Single Pulse Avalanche Energy (3)		E_{AS}	230	mJ
Single Pulse Avalanche Current (L= 0.1mH)		I_{AS}	38	A
Power Dissipation	T_C = 25°C	P_D	94	W
	T_C = 100°C		47	W
Junction & Storage Temperature Range		T_J, T_{STG}	-55 ~ +175	°C

Thermal Characteristics

Parameter	Symbol	Typ.	Max.	Unit
Thermal Resistance, Junction-to-Ambient (4)	$R_{\theta JA}$	37	46	°C/W
Thermal Resistance, Junction-to-Case (5)	$R_{\theta JC}$	1.2	1.6	°C/W

Electrical Characteristics (@ T_J = 25°C, unless otherwise specified.)

Parameter	Symbol	Test Condition	Min.	Typ.	Max.	Unit
Off Characteristics [6]						
Drain-Source Breakdown Voltage	$V_{(BR)DSS}$	V_{GS} = 0V, I_D = 250μA	40	-	-	V
Zero Gate Voltage Drain Current	I_{DSS}	V_{DS} = 40V, V_{GS} = 0V	-	-	1.0	μA
		T_J = 125°C	-	-	100	μA
Gate-Source Leakage Current	I_{GSS}	V_{GS} = ±20V, V_{DS} = 0V	-	-	±100	nA
On Characteristics [6]						
Gate Threshold Voltage	$V_{GS(th)}$	V_{GS} = V_{DS}, I_D = 250μA	1.2	1.7	2.5	V
Static Drain-Source On-Resistance	$R_{DS(ON)}$	V_{GS} = 10V, I_D = 20A	-	2.0	2.4	mΩ
		V_{GS} = 4.5V, I_D = 15A	-	2.9	3.7	mΩ
Forward Transconductance	g_{fs}	V_{DS} = 5.0V, I_D = 20A	-	40	-	S
Diodes Forward Voltage	V_{SD}	I_S = 2.0A, V_{GS} = 0V	-	0.7	1.2	V
Dynamic Characteristics [7]						
Input Capacitance	C_{iss}	V_{DS} = 20V, V_{GS} = 0V, f = 1MHz	-	1602	2083	pF
Output Capacitance	C_{oss}		-	853	1109	pF
Reverse Transfer Capacitance	C_{rss}		-	46	92	pF
Gate Resistance	R_g	V_{GS} = 0V, V_{DS} = 0V, f = 1MHz	-	2.6	-	Ω
Switching Characteristics [7]						
Turn-On DelayTime	$t_{d(on)}$	V_{GS} = 10V, V_{DS} = 20V I_D = 20A, R_{GEN} = 3.0Ω	-	2.8	-	ns
Rise Time	t_r		-	3.2	-	ns
Turn-Off DelayTime	$t_{d(off)}$		-	25	-	ns
Fall Time	t_f		-	13	-	ns
Gate Charge Characteristics [7]						
Total Gate Charge (V_{GS} = 10V)	Q_g	V_{DS} = 20V, I_D = 20A V_{GS} = 10V	-	24	31	nC
Total Gate Charge (V_{GS} = 4.5V)	Q_g		-	11.6	15.0	nC
Gate-Source Charge	Q_{gs}		-	4.5	6.8	nC
Gate-Drain Charge	Q_{gd}		-	3.6	5.4	nC
Gate Plateau Voltage	$V_{plateau}$		-	2.9	-	V
Drain-Source Diode Characteristics [7]						
Body Diode Reverse Recovery Time	t_{rr}	I_F = 20A, dI/dt = 100A/μs, T_J = 25°C	-	35	-	ns
Body Diode Reverse Recovery Charge	Q_{rr}		-	21	-	nC
Diode Forward Current	I_S	T_C = 25°C	-	-	146	A

Notes:

1. This current is chip limited, which is calculated based on Rthjc.
2. This current is calculated on single pulse with 10μs Single Pulse.
3. Defined by design, not subject to production test, E_{AS} condition: T_J=25°C, V_{DD}=20V, V_{GS}=10V, L=1.0mH.
4. Device mounted on FR-4 substrate PC board with 2oz copper in 1inch square cooling area.
5. Thermal resistance from junction to the exposed pad.
6. Short duration pulse test used to minimize self-heating effect.
7. Defined by design, not subject to production.

Typical Electrical and Thermal Characteristics

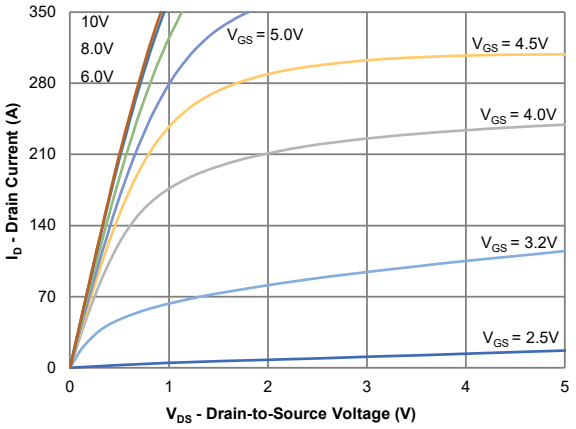


Figure 1: Output Characteristics

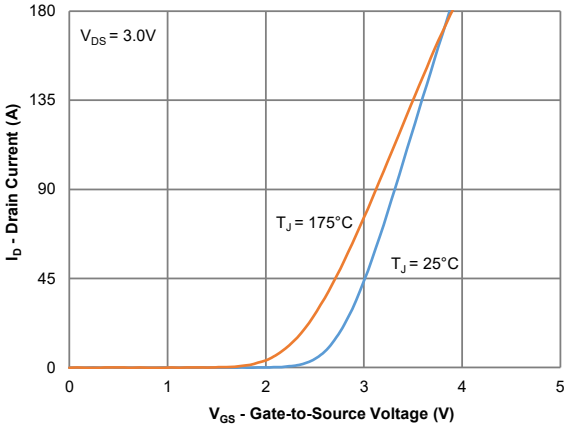


Figure 2: Transfer Characteristics

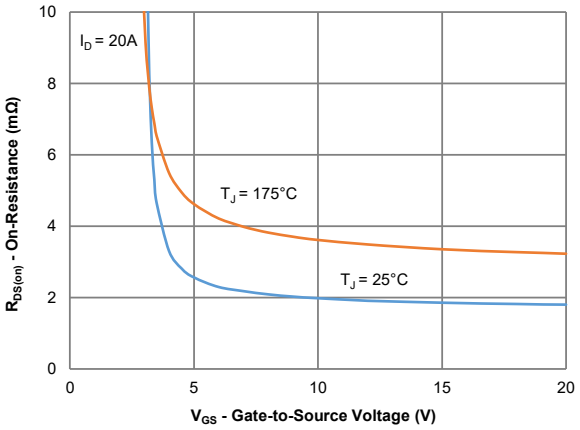


Figure 3: On-Resistance vs. Gate-Source Voltage

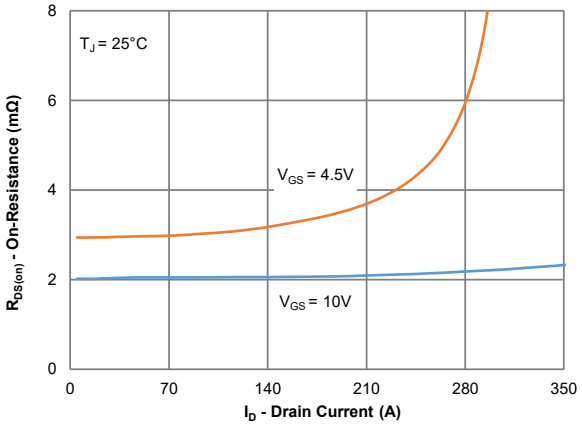


Figure 4: On-Resistance vs. Drain Current and Gate Voltage

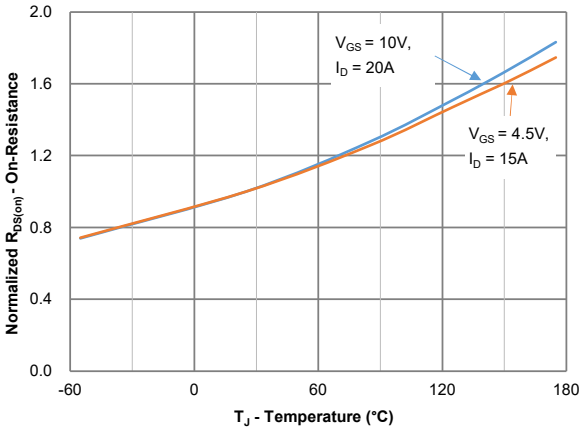


Figure 5: On-Resistance vs. Junction Temperature

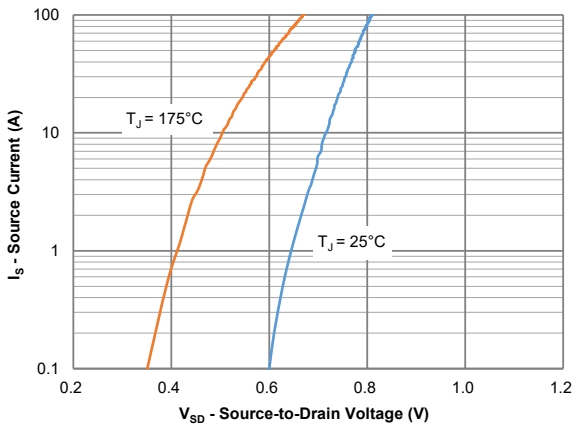


Figure 6: Source-Drain Diode Forward Voltage

Typical Electrical and Thermal Characteristics

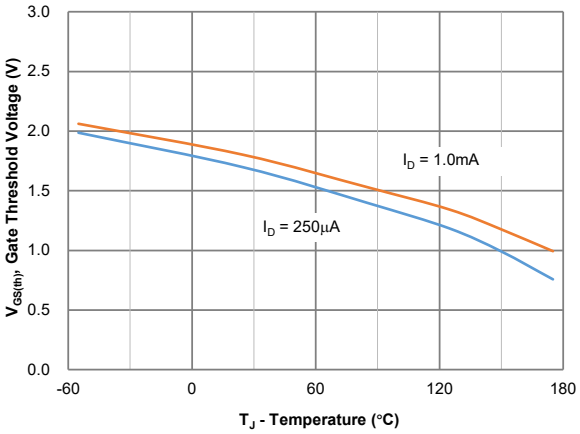


Figure 7: Gate Threshold Variation vs. Junction Temperature

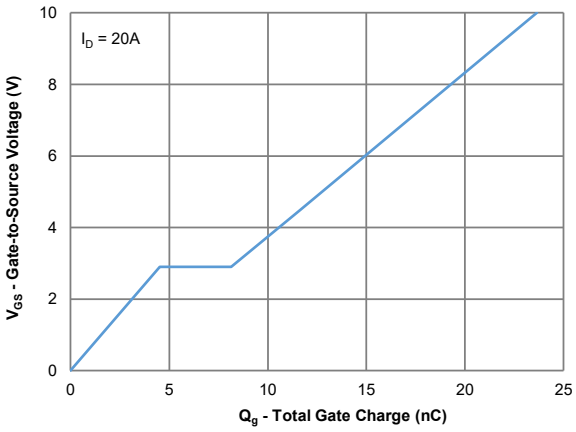


Figure 8: Gate Charge Characteristics

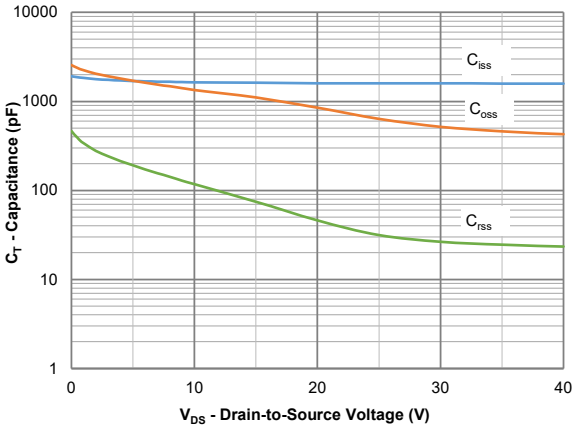


Figure 9: Capacitance Characteristics

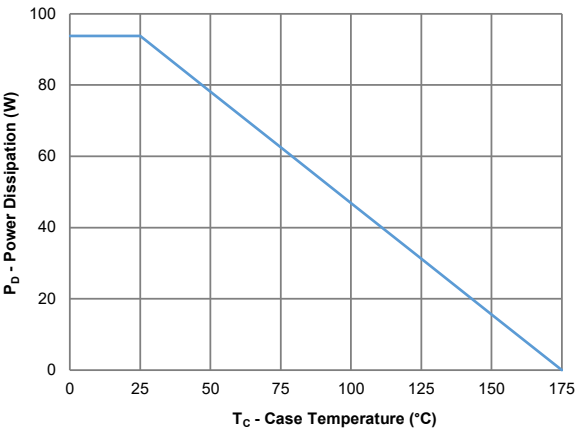


Figure 10: Power Derating

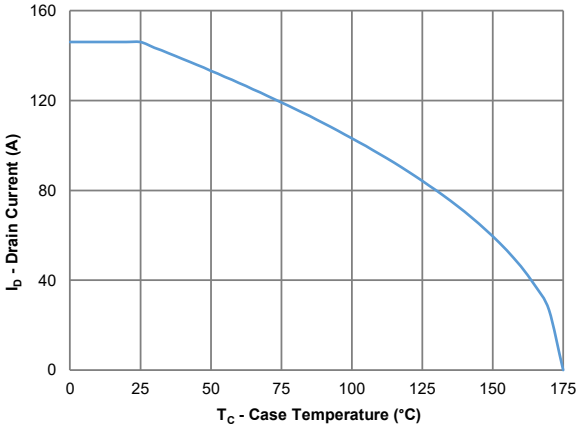


Figure 11: Current Derating

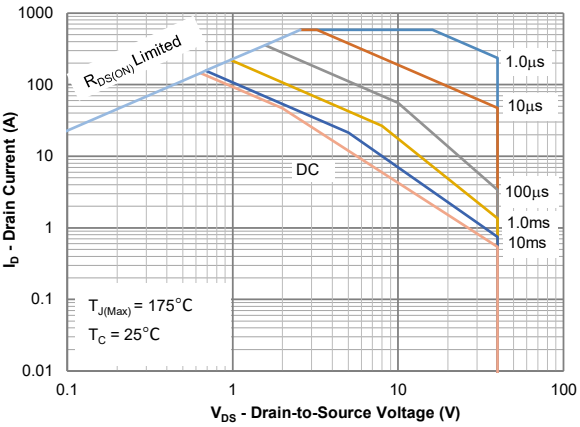


Figure 12: Safe Operating Area

Typical Electrical and Thermal Characteristics

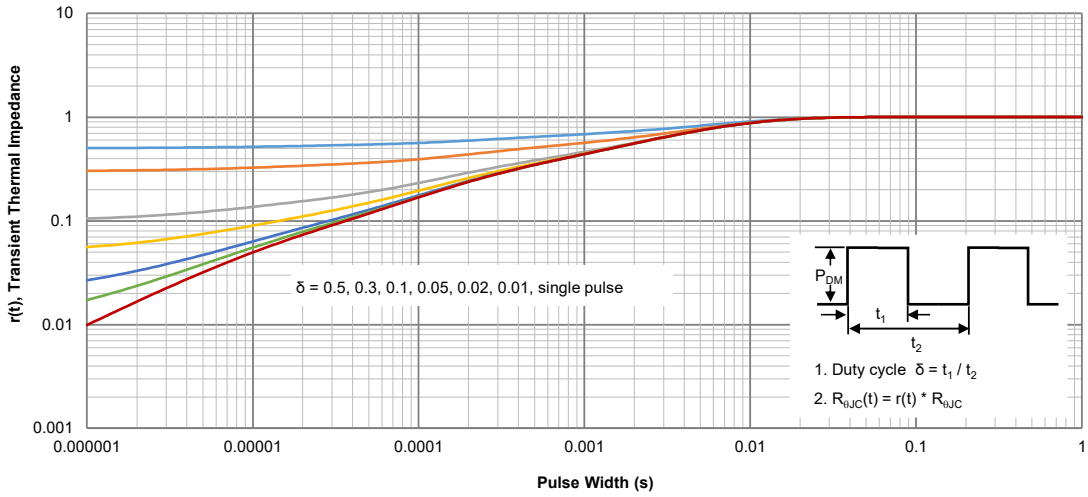
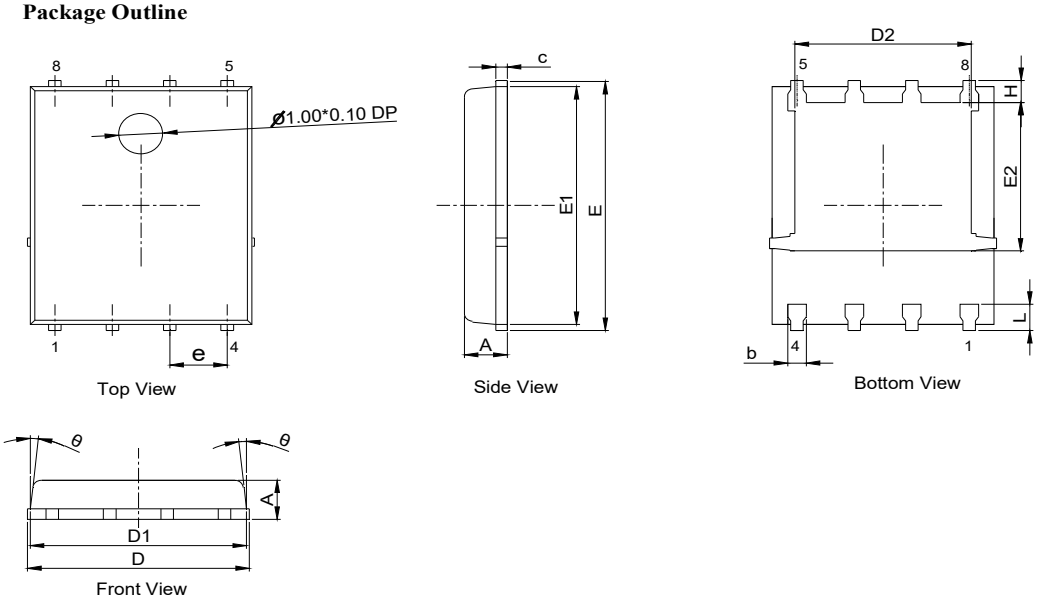


Figure 13: Normalized Maximum Transient Thermal Impedance

PDFN5060-8L Package Outline

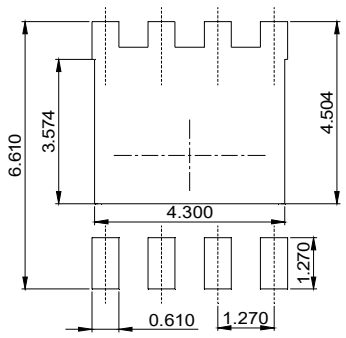
Package Outline



NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M,1994.
2. ALL DIMNESIONS IN MILLIMETER (ANNGLE IN DEGREE).
3. DIMENSIONS D1 AND E1 DO NOT INCLUDE MOLD FLASH PROTRUSIONS OR GATE BURRS.

DIM.	MILLIMETER		
	MIN.	NOM.	MAX.
A	0.90	1.05	1.20
b	0.30	0.41	0.51
c	0.20	0.25	0.34
D	4.80	--	5.40
D1	4.65	--	5.30
D2	3.60	--	4.25
E	5.90	--	6.50
E1	5.70	--	6.00
E2	3.38	--	3.92
e	1.27BSC		
H	0.40	--	0.93
L	0.40	--	0.91
θ	0°	--	12°

Recommended Soldering Footprint



DIMENSIONS:MILLIMETERS